AQ10.3: Activity Questions 3 - Not Graded

This assignment will not be graded and is only for practice.
Consider the following functions

$$f_1(x, y) \text{f1(x, y)= } 5\sqrt{x^2 + y^2} 5 \quad \text{x2 + y2}$$

$$f_2(x, y, z) \text{f2(x, y, z)} = x^2 z^2 + yz + x^2 y^2 z^2 \text{x2z2 + yz + x2y2z2}$$

$$f_3(x, y) \text{f3(x, y)= } \begin{cases} y \sin\frac{1}{y} + x & y \neq 0 \\ 0 & y = 0. \end{cases} \{\text{y sin y1 + xy0if y} \quad = 0 \text{if y = 0.}$$

If tangent plane exists at the origin for f_i fi, then enter the value of ii.

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 2
1 point
1 point
Which of the following equations represents the tangent plane to the function $f(x, y) = 3x^2 + xy - 2y^2$ f(x, y) = 3x2 + xy − 2y2 at the point (1,-1)

- ○ $5x + 5y = 0$5x + 5y = 0.
- ○ $5x + 5y - z = 0$5x + 5y − z = 0.
- ○ $x + y - z = 0$x + y − z = 0.
- ○ $5x + 5y + z = 0$5x + 5y + z = 0.

No, the answer is incorrect.
Score: 0
Accepted Answers:
$5x + 5y - z = 0$5x + 5y − z = 0.
1 point
Consider a function $f(x, y) = \frac{1}{2}(xy - \frac{x}{y})$f(x, y) = 21(xy − yx). Let TT be the tangent plane at point (1,2). Which of the following points lie on the tangent plane TT?

- ☐ $(0,0,-\frac{5}{4}-45)$
- ☐ $(1,0,\frac{1}{4}41)$
- ☐ $(2, -4, -\frac{9}{4}-49)$
- ☐ (1, 2, -2)

No, the answer is incorrect.
Score: 0

Accepted Answers:
$(0,0,-\frac{5}{4}-45)$
$(2, -4, -\frac{9}{4}-49)$

1 point

Consider a function $f(x, y) = xy$f(x, y) = xy. Let T_1 T1 be the tangent plane at point (1,1), T_2 T2 be the tangent plane at point (1,-1). Then which of the following option is true?

- ○ T_1, T_2 T1, T2 intersect in only one point.
- ○ T_1, T_2 T1, T2 intersect in infinitely many points.
- ○ T_1, T_2 T1, T2 do not intersect.
- ○ None of the above

No, the answer is incorrect.
Score: 0
Accepted Answers:
T_1, T_2 T1, T2 intersect in infinitely many points.

1 point

Let f f be a function from R^2 to$R$R2 to R and TT be the tangent plane at a point. Then which of the following option(s) is (are) true.

- ☐ TT is an affine subspace of the vector space R^3 R3 with respect to usual addition and scalar multiplication.
- ☐ TT is never an affine subspace of the vector space R^3 R3 with respect to usual addition and scalar multiplication.
- ☐ TT can be a subspace of the vector space R^3 R3 with respect to usual addition and scalar multiplication.
- ☐ None of the above.

No, the answer is incorrect.
Score: 0
Accepted Answers:
TT is an affine subspace of the vector space R^3 R3 with respect to usual addition and scalar multiplication.
TT can be a subspace of the vector space R^3 R3 with respect to usual addition and scalar multiplication.

1 point

Consider a function $f(x, y, z) = e^{x+y}$ sin2zf(x, y, z) = ex+y sin 2z. Which of the followings is the tangent (hyper) plane at the point $(1, 0, \frac{\pi}{4})(1, 0, 4\pi)$?

- ○ $u = ex + ey$u = ex + ey
- ○ $u = ex + ey + z$u = ex + ey + z
- ○ $u = ex + ey + ez$u = ex + ey + ez
- ○ None of the above.

No, the answer is incorrect.
Score: 0
Accepted Answers:
$u = ex + ey$u = ex + ey

1 point

Let $f: R^3 \rightarrow R$f : R3 → R be a function and L_f Lf be the linear approximation of ff at point (a, b) (a, b). Which of the following options is true?

○ L_f Lf is a linear function.

○ L_f Lf is a quadratic function.

○ L_f Lf is a logarithmic function.

○ None of the above.

No, the answer is incorrect.

Score: 0

Accepted Answers:

L_f Lf is a linear function.

1 point

Consider a function $f: R^2 \rightarrow R$f : R2 → R defined as $f(x, y) = xy\tan(x)\sin^{-1}(y + \frac{\sqrt{3}}{2}) + 1$ f(x, y) = xy tan(x) sin−1(y + 2 3 √) + 1. Which of the following is the linear approximation of ff at the point $(\frac{\pi}{4}, 0)(4\pi, 0)$

○ $L_f(x, y) = x + \frac{\pi^2}{12}y$Lf(x, y) = x + 12π2y

○ $L_f(x, y) = 1 + \frac{\pi^2}{12}y$Lf(x, y) = 1 + 12π2y

○ $L_f(x, y) = 1 + y$Lf(x, y) = 1 + y

○ $L_f(x, y) = \frac{\pi^2}{12}y$Lf(x, y) = 12π2y

No, the answer is incorrect.

Score: 0

Accepted Answers:

$L_f(x, y) = 1 + \frac{\pi^2}{12}y$Lf(x, y) = 1 + 12π2y

Consider the three functions $f_1 = x^2 + y^2, f_2 = xy$f1 = x2 + y2, f2 = xy and $f_3 = x + y$f3 = x + y. Let T_1, T_2 T1, T2 and T_3 T3 be the tangent planes at points $(-\frac{1}{2}, \frac{1}{2})(-21, 21)$ of f_1 f1, (-1, 1,) of f_2 f2 and (0,0) of f_3 f3 respectively. If (a, b, c)(a, b, c) are the intersection point of the T_1, T_2 T1, T2 and T_3 T3, then find the value of $a + 2b + c$a + 2b + c.

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 1

1 point

1 point

Which of the following is the linear approximation of the function $f(x, y) = e^x \sin y + xe^x$ f(x, y) = ex sin y + xex at the point (0, 0)

○ $L_f(0, 0) = 2x + y$Lf(0, 0) = 2x + y

○ $L_f(0, 0) = x$Lf(0, 0) = x

○

$L_f(0,0) = 2x + 2y$Lf(0, 0) = 2x + 2y

○

$L_f(0,0) = x + y$Lf(0, 0) = x + y

No, the answer is incorrect.

Score: 0

Accepted Answers:

$L_f(0,0) = x + y$Lf(0, 0) = x + y

1 point

Production of number of breads depends upon the quantity of raw material which includes egg, wheat and sugar . xx denotes the number of eggs, yy denotes the quantity of wheat (in kg) and zz denotes the quantity of sugar (in kg) and defined as $f(x,y,z) = xyz$f(x, y, z) = xyz. Quality of the bread is measured by the value of linear approximation ($L_f(x,y,z)$)(Lf(x, y, z)) with respect to 1 egg, 1 kg wheat, and 1 kg sugar. If value of the linear approximation is greater than 10, then quality of product is not good and if less than 10 then quality of the product is good.

Use this information to answer question.

Which of the following option(s) is (are) true?

☐ If 2 eggs, 5 units wheat, 3 units sugar is used to produce breads, then the quality of produced breads is good.

☐ If 5 eggs, 10 units wheat, 3 units sugar is used to produce breads, then the quality of produced breads is not good.

☐ If 7 eggs, 8 units wheat, 9 units sugar is used to produce breads, then the quality of produced breads is good.

☐ If 5 eggs, 3 units wheat, 3 units sugar is used to produce breads, then the quality of produced breads is not good.

No, the answer is incorrect.

Score: 0

Accepted Answers:

If 2 eggs, 5 units wheat, 3 units sugar is used to produce breads, then the quality of produced breads is good.

If 5 eggs, 10 units wheat, 3 units sugar is used to produce breads, then the quality of produced breads is not good.

Consider a functions $f: R^3 \rightarrow R$f : R3 → R defined as: $f(x,y,z) = x^2 + y^2 + z^2$.
f(x, y, z) = x2 + y2 + z2. If $L_f(x,y,z)$Lf(x, y, z) is the linear approximation of $f(x,y,z)$f(x, y, z) at point (1, 1, 0), then find the value of $\nabla L_f(x,y,z) \cdot (3,2,1)$∇Lf(x, y, z) · (3, 2, 1).

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 10

1 point

1 point

Consider a function $f(x,y,z) = ax + by + cz$f(x, y, z) = ax + by + cz, where $a, b, c \in R$
a, b, c ∈ R. If $L_f(x,y,z)$Lf(x, y, z) is the linear approximation of $f(x,y,z)$f(x, y, z) at the origin, then which of the following options is true?

○

$L_f(x,y,z) = 2f(x,y,z)$Lf(x, y, z) = 2f(x, y, z)

○

$L_f(x,y,z) = -f(x,y,z)$Lf(x, y, z) = −f(x, y, z)

○

$L_f(x,y,z) = f(x,y,z)$Lf(x, y, z) = f(x, y, z)

$L_f\,(x,\,y,\,z) = 0$Lf (x, y, z) = 0

No, the answer is incorrect.

Score: 0

Accepted Answers:

$L_f\,(x,\,y,\,z) = f\,(x,\,y,\,z)$Lf (x, y, z) = f (x, y, z)

Check Answers

Your score is: 0/13